

```
Setting environment for using XAMPP for Windows.
23301585@CSE-12F32L-L31 c:\xampp
# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.4.22-MariaDB mariadb.org binary distribution

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
MariaDB [(none)]> CREATE DATABASE Smart_Navigation_db
-> ;
Query OK, 1 row affected (0.004 sec)
```

```
MariaDB [(none)]> USE Smart_Navigation_db;
Database changed
MariaDB [Smart_Navigation_db]> CREATE TABLE USER (
->   user_id INT AUTO_INCREMENT PRIMARY KEY,
->   name VARCHAR(100) NOT NULL,
->   email VARCHAR(100) UNIQUE NOT NULL,
->   password VARCHAR(255) NOT NULL,
->   preferred_budget DECIMAL(10,2),
->   preferred_time INT, -- stored in minutes
->   comfort_level INT -- 1 to 5 scale
-> );
Query OK, 0 rows affected (0.046 sec)
```

```
MariaDB [Smart_Navigation_db]> CREATE TABLE LOCATION (
->   location_id INT AUTO_INCREMENT PRIMARY KEY,
->   location_name VARCHAR(255) NOT NULL,
->   latitude DECIMAL(10, 8) NOT NULL,
->   longitude DECIMAL(11, 8) NOT NULL,
->   area_zone VARCHAR(50)
-> );
Query OK, 0 rows affected (0.048 sec)
```

```
MariaDB [Smart_Navigation_db]> CREATE TABLE TRANSPORT_MODE (
->   transport_id INT AUTO_INCREMENT PRIMARY KEY,
->   transport_type VARCHAR(50) NOT NULL,
->   average_speed DECIMAL(5,2), -- km/h
->   base_fare DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.057 sec)
```

```
MariaDB [Smart_Navigation_db]> CREATE TABLE ROUTE (
->   route_id INT AUTO_INCREMENT PRIMARY KEY,
->   source_location_id INT,
->   destination_location_id INT,
->   total_distance DECIMAL(10,2), -- in km
->   estimated_time INT, -- in minutes
->   estimated_cost DECIMAL(10,2),
->   FOREIGN KEY (source_location_id) REFERENCES LOCATION(location_id),
->   FOREIGN KEY (destination_location_id) REFERENCES LOCATION(location_id)
-> );
Query OK, 0 rows affected (0.049 sec)
```

```

MariaDB [Smart_Navigation_db]> CREATE TABLE ROUTE_SEGMENT (
->   segment_id INT AUTO_INCREMENT PRIMARY KEY,
->   route_id INT,
->   transport_id INT,
->   start_location_id INT,
->   end_location_id INT,
->   segment_distance DECIMAL(10,2),
->   segment_time INT,
->   segment_cost DECIMAL(10,2),
->   FOREIGN KEY (route_id) REFERENCES ROUTE(route_id),
->   FOREIGN KEY (transport_id) REFERENCES TRANSPORT_MODE(transport_id),
->   FOREIGN KEY (start_location_id) REFERENCES LOCATION(location_id),
->   FOREIGN KEY (end_location_id) REFERENCES LOCATION(location_id)
-> );

```

Query OK, 0 rows affected (0.056 sec)

```

MariaDB [Smart_Navigation_db]> CREATE TABLE TRIP_HISTORY (
->   trip_id INT AUTO_INCREMENT PRIMARY KEY,
->   user_id INT,
->   route_id INT,
->   travel_time INT,
->   travel_cost DECIMAL(10,2),
->   trip_date DATETIME,
->   FOREIGN KEY (user_id) REFERENCES USER(user_id),
->   FOREIGN KEY (route_id) REFERENCES ROUTE(route_id)
-> );

```

Query OK, 0 rows affected (0.058 sec)

```

MariaDB [Smart_Navigation_db]> CREATE TABLE INCIDENT_REPORT (
->   incident_id INT AUTO_INCREMENT PRIMARY KEY,
->   user_id INT,
->   location_id INT,
->   incident_type VARCHAR(50), -- e.g., 'Flood', 'Accident', 'Protest'
->   severity VARCHAR(20), -- e.g., 'Low', 'Medium', 'High'
->   timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
->   status VARCHAR(20) DEFAULT 'Active',
->   FOREIGN KEY (user_id) REFERENCES USER(user_id),
->   FOREIGN KEY (location_id) REFERENCES LOCATION(location_id)
-> );

```

Query OK, 0 rows affected (0.056 sec)

```

MariaDB [Smart_Navigation_db]> CREATE TABLE TRAFFIC_DATA (
->   traffic_id INT AUTO_INCREMENT PRIMARY KEY,
->   location_id INT,
->   congestion_level VARCHAR(20), -- e.g., 'Clear', 'Heavy', 'Gridlock'
->   avg_speed DECIMAL(5,2),
->   date DATE,
->   time_slot TIME,
->   FOREIGN KEY (location_id) REFERENCES LOCATION(location_id)
-> );

```

Query OK, 0 rows affected (0.049 sec)

```

MariaDB [Smart_Navigation_db]> CREATE TABLE TRANSPORT_REVIEWS (
->   review_id INT AUTO_INCREMENT PRIMARY KEY,
->   user_id INT,
->   transport_id INT,
->   rating INT CHECK (rating >= 1 AND rating <= 5),
->   comment TEXT,
->   timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
->   FOREIGN KEY (user_id) REFERENCES USER(user_id),
->   FOREIGN KEY (transport_id) REFERENCES TRANSPORT_MODE(transport_id)
-> );

```

Query OK, 0 rows affected (0.050 sec)

```

MariaDB [Smart_Navigation_db]> CREATE TABLE CACHED_ROUTES (
->   cache_id INT AUTO_INCREMENT PRIMARY KEY,
->   source_location_id INT,
->   destination_location_id INT,
->   route_data JSON, -- Stores the compiled route steps
->   calculated_time DATETIME DEFAULT CURRENT_TIMESTAMP,
->   FOREIGN KEY (source_location_id) REFERENCES LOCATION(location_id),
->   FOREIGN KEY (destination_location_id) REFERENCES LOCATION(location_id)
-> );

```

Query OK, 0 rows affected (0.052 sec)

```

MariaDB [Smart_Navigation_db]> CREATE TABLE TRANSPORT_FARE (
->   fare_id          INT AUTO_INCREMENT PRIMARY KEY,
->   transport_id     INT,
->   base_fare        DECIMAL(10,2),
->   cost_per_km      DECIMAL(8,4),
->   time_multiplier  DECIMAL(5,2) DEFAULT 1.0,
->   area_zone        VARCHAR(50),
->   FOREIGN KEY (transport_id) REFERENCES TRANSPORT_MODE(transport_id)
-> );
Query OK, 0 rows affected (0.059 sec)

MariaDB [Smart_Navigation_db]> CREATE TABLE TRANSPORT_AVAILABILITY (
->   availability_id  INT AUTO_INCREMENT PRIMARY KEY,
->   transport_id     INT,
->   location_id      INT,
->   available_count  INT DEFAULT 0,
->   timestamp        DATETIME DEFAULT CURRENT_TIMESTAMP,
->   FOREIGN KEY (transport_id) REFERENCES TRANSPORT_MODE(transport_id),
->   FOREIGN KEY (location_id) REFERENCES LOCATION(location_id)
-> );
Query OK, 0 rows affected (0.059 sec)

```

A few Data(Value) Insertion

```

MariaDB [Smart_Navigation_db]> INSERT INTO USER (name, email, password, preferred_budget, preferred_time, comfort_level)
-> VALUES ('Ahmed', 'ahmed@example.com', 'password123', 200.00, 60, 4);
Query OK, 1 row affected (0.052 sec)

MariaDB [Smart_Navigation_db]> SELECT * FROM USER;
+-----+-----+-----+-----+-----+-----+-----+
| user_id | name  | email          | password | preferred_budget | preferred_time | comfort_level |
+-----+-----+-----+-----+-----+-----+-----+
| 1       | Ahmed | ahmed@example.com | password123 | 200.00          | 60             | 4             |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)

```

```

MariaDB [Smart_Navigation_db]> INSERT INTO LOCATION (location_name, latitude, longitude, area_zone) VALUES
-> ('Mirpur 10', 23.8052, 90.3696, 'North'),
-> ('Banani', 23.7940, 90.4043, 'North'),
-> ('Gulshan 1', 23.7806, 90.4162, 'Central'),
-> ('Dhanmondi 27', 23.7533, 90.3768, 'Central'),
-> ('Motijheel', 23.7289, 90.4189, 'South');
Query OK, 5 rows affected (0.011 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [Smart_Navigation_db]> SELECT * FROM LOCATION;
+-----+-----+-----+-----+-----+
| location_id | location_name | latitude | longitude | area_zone |
+-----+-----+-----+-----+-----+
| 1           | Mirpur 10     | 23.80520000 | 90.36960000 | North     |
| 2           | Banani        | 23.79400000 | 90.40430000 | North     |
| 3           | Gulshan 1     | 23.78060000 | 90.41620000 | Central   |
| 4           | Dhanmondi 27  | 23.75330000 | 90.37680000 | Central   |
| 5           | Motijheel     | 23.72890000 | 90.41890000 | South     |
+-----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

```

```
MariaDB [Smart_Navigation_db]> INSERT INTO TRANSPORT_MODE (transport_type, average_speed, base_fare) VALUES
-> ('Bus', 20.00, 20.00),
-> ('Metro Rail', 45.00, 40.00),
-> ('Rickshaw', 12.00, 30.00);
Query OK, 3 rows affected (0.010 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
MariaDB [Smart_Navigation_db]> SELECT * FROM TRANSPORT_MODE;
+-----+-----+-----+-----+
| transport_id | transport_type | average_speed | base_fare |
+-----+-----+-----+-----+
| 1 | Bus | 20.00 | 20.00 |
| 2 | Metro Rail | 45.00 | 40.00 |
| 3 | Rickshaw | 12.00 | 30.00 |
+-----+-----+-----+-----+
3 rows in set (0.000 sec)
```

```
MariaDB [Smart_Navigation_db]> INSERT INTO ROUTE (source_location_id, destination_location_id, total_distance, estimated_time, estimated_cost) VALUES
-> (1, 5, 12.5, 45, 120.00),
-> (1, 5, 13.0, 60, 50.00),
-> (1, 5, 12.0, 30, 200.00);
Query OK, 3 rows affected (0.012 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
MariaDB [Smart_Navigation_db]> SELECT * FROM ROUTE;
+-----+-----+-----+-----+-----+-----+
| route_id | source_location_id | destination_location_id | total_distance | estimated_time | estimated_cost |
+-----+-----+-----+-----+-----+-----+
| 1 | 1 | 5 | 12.50 | 45 | 120.00 |
| 2 | 1 | 5 | 13.00 | 60 | 50.00 |
| 3 | 1 | 5 | 12.00 | 30 | 200.00 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.000 sec)
```

```
MariaDB [Smart_Navigation_db]> INSERT INTO TRAFFIC_DATA (location_id, congestion_level, avg_speed, date, time_slot) VALUES
-> (1, 'Heavy', 10.5, '2026-04-19', '17:15:00'),
-> (1, 'Heavy', 9.0, '2026-04-19', '17:45:00'),
-> (2, 'Clear', 35.0, '2026-04-19', '17:30:00');
Query OK, 3 rows affected (0.010 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
MariaDB [Smart_Navigation_db]> SELECT * FROM TRAFFIC_DATA;
+-----+-----+-----+-----+-----+-----+
| traffic_id | location_id | congestion_level | avg_speed | date | time_slot |
+-----+-----+-----+-----+-----+-----+
| 1 | 1 | Heavy | 10.50 | 2026-04-19 | 17:15:00 |
| 2 | 1 | Heavy | 9.00 | 2026-04-19 | 17:45:00 |
| 3 | 2 | Clear | 35.00 | 2026-04-19 | 17:30:00 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.000 sec)
```

```
MariaDB [Smart_Navigation_db]> INSERT INTO INCIDENT_REPORT (user_id, location_id, incident_type, severity, timestamp, status) VALUES
-> (1, 3, 'Waterlogging', 'High', '2026-04-19 16:00:00', 'Active');
Query OK, 1 row affected (0.011 sec)
```

```
MariaDB [Smart_Navigation_db]> SELECT * FROM INCIDENT_REPORT;
+-----+-----+-----+-----+-----+-----+
| incident_id | user_id | location_id | incident_type | severity | timestamp | status |
+-----+-----+-----+-----+-----+-----+
| 1 | 1 | 3 | Waterlogging | High | 2026-04-19 16:00:00 | Active |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```